

A Continuous-Time SOC-Based Battery Drain Model for Smartphone Time-to-Empty Prediction with Thermal Feedback

Summary

Smartphone battery life exhibits substantial variability even when devices start from similar initial charge levels. This phenomenon arises from time-varying user behavior, background system processes, network activity, and thermal effects, which jointly introduce strong nonlinearities into battery discharge dynamics. Traditional linear or static power-based models are therefore insufficient to explain why certain usage scenarios lead to disproportionately rapid battery depletion.

In this study, we develop a continuous-time analytical framework to characterize the evolution of battery state of charge (SOC) and to estimate time-to-empty (TTE) and capture nonlinear battery behavior under diverse usage scenarios. We propose **continuous-time SOC model with thermal feedback (CT-SOC-TF Model)**, a coupled system of differential equations that integrates scenario-dependent power decomposition, temperature-dependent effective capacity, thermal throttling feedback, network tail energy dynamics, and internal resistance losses. The key innovation lies in **explicitly modeling bidirectional thermal feedback**, whereby power consumption raises temperature while elevated temperature suppresses power and accelerates effective energy loss. The SOC trajectory is obtained by numerically solving the coupled SOC–thermal system forward in time, enabling direct estimation of TTE under both stationary and composite usage patterns.

Simulation results of TTE across five representative smartphone scenarios, ranging from **334.5 to 1002.4 mins with 100% initial SOC**, **213.7 to 642.6 mins with 66% initial SOC** and **96.7 to 294.0 mins with 33% initial SOC** show that the model consistently explains differences in battery lifetime through usage intensity and thermal response. Mobile gaming yields the shortest TTE, while music playback produces the slowest depletion, with the relative ordering remaining stable across initial SOC levels. Quantitative evaluation against observed SOC traces yields **moderate prediction errors (4.5–5.1%)**, consistent with an uncalibrated phenomenological model. Overall, the CT-SOC-TF Model provides a mechanism-valid and interpretable framework for understanding battery drain, identifying dominant drivers of rapid depletion, and informing practical battery-aware usage and system-level strategies.

Overall, the CT-SOC-TF Model provides a mechanism-consistent and interpretable framework for understanding smartphone battery drain. It highlights the dominant drivers of rapid depletion and offers useful insights for battery-aware user behavior and system-level energy management strategies. An executive-style summary can be found in section 9.

Keywords: Continuous -Time equation Thermal Feedback Mechanism Pareto distribution Arrhenius equation

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1 Introduction

1.1 Problem Background

Battery life remains a fundamental constraint in smartphone usage. Despite ongoing improvements in hardware performance, users often observe substantial variation in device operating time even when starting from similar charge levels, indicating that battery depletion cannot be explained by usage duration alone.

Such variability stems from the time-varying and multifaceted nature of smartphone energy demand. Power consumption fluctuates continuously as user activities, background processes, network communication, and environmental conditions change. In particular, short periods of intensive usage may have a disproportionate impact on battery life, while extended low-power operation may result in comparatively slow depletion, complicating both interpretation and prediction of battery behavior.

To address this challenge, a coherent modeling framework is required to link time-dependent power demand to the evolution of battery state of charge (SOC). By treating battery depletion as a continuous dynamical process, it becomes possible to systematically relate usage patterns to SOC trajectories and remaining operating time, which motivates the modeling approach developed in the following sections.



Figure 1: Cellphone Charging

1.2 Problem Restatement and Modeling Objectives

We aim to develop an analytical framework for understanding and predicting smartphone battery behavior under realistic usage conditions. The modeling objectives are threefold:

1. Task1: Continuous SOC Modeling

Develop a continuous-time mathematical model that captures the evolution of battery state of charge (SOC) as a function of time, incorporating major drivers of power consumption such as screen usage, processor workload, network activity, and background processes.

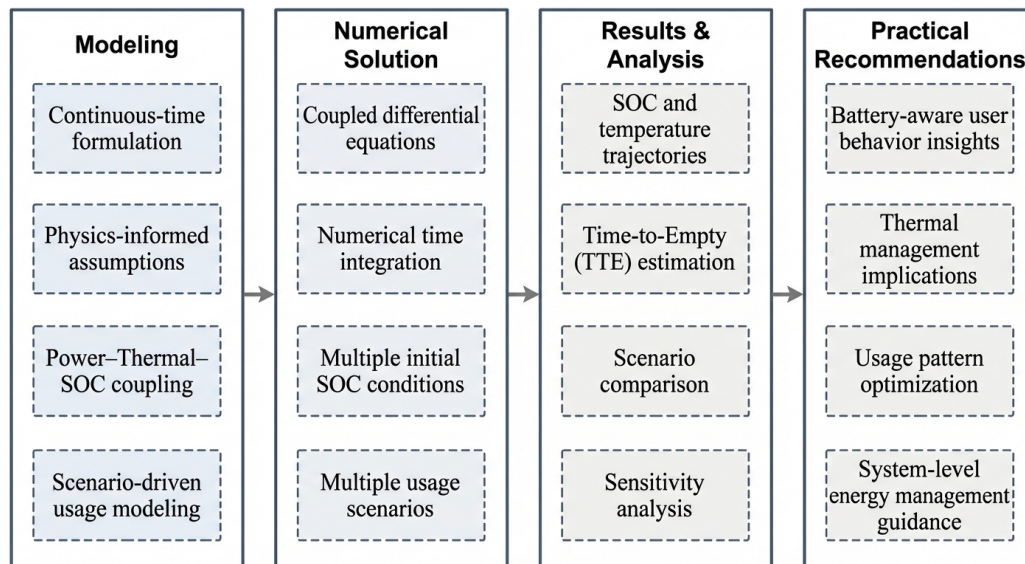
2. Task2: Time-to-Empty Estimation

Use the proposed model to estimate time-to-empty (TTE) under different initial SOC levels and usage scenarios, and to identify usage patterns that lead to rapid versus gradual battery depletion.

3. Task3: Sensitivity and Practical Insights

Examine the sensitivity of SOC and TTE predictions to modeling assumptions, parameters, and usage variability, and translate the resulting insights into practical recommendations for battery-aware user behavior and system-level strategies.

1.3 Our work



2 Assumptions and Justifications

- **Assumption 1:** In Tasks 1 and 2, the battery state of health (SOH) is assumed to be constant and equal to 1.

Justification: Tasks 1 and 2 consider short-term battery discharge following a single charging cycle, during which battery aging effects are negligible. Since SOH evolves on a much longer time scale than SOC and temperature, it is treated as constant to focus the model on usage-driven and thermal dynamics. The effect of relaxing this assumption is examined in the sensitivity analysis.

- **Assumption 2:** In Task 2, each usage scenario is assumed to follow a stationary operating pattern, such that the corresponding control inputs exhibit a unique and time-invariant trend and can be extrapolated from finite-duration observations.

Justification: Each scenario represents a sustained and representative usage mode whose statistical characteristics remain stable over time. This assumption allows finite-duration observations to be extrapolated for long-term SOC and time-to-empty estimation while preserving the tractability of the continuous-time model. Non-stationary usage patterns are later examined through composite scenario analysis.

- **Assumption 3:** Model parameters are assumed to be scenario-invariant and device-independent, and are selected to represent typical smartphone behavior rather than any specific device.

Justification: This study aims to develop a phenomenological and interpretable model that captures general battery behavior across usage scenarios rather than device-specific numerical accuracy. Assuming scenario-invariant and device-independent parameters avoids overfitting and highlights the structural relationships among power consumption, temperature, and SOC. The implications of this assumption are discussed in the model evaluation and limitations.

3 Notations

Symbol	Definition
t	Time (s)
$s(t)$ or $\text{SOC}(t)$	Battery state of charge (%)
SOC_0	Initial state of charge (%)
$P_{\text{sys}}(t)$	System power consumption (W)
$P_{\text{loss}}(t)$	Power loss due to internal resistance (W)
$P_{\text{heat}}(t)$	Internal heat generation (W)
$T(t)$	Battery (core) temperature (K)
T_{env}	Ambient temperature (K)
$\eta_{\text{temp}}(T)$	Temperature-dependent capacity factor
$\phi(T)$	Thermal throttling factor
$E_{\text{cap},J}(T, H)$	Effective usable battery capacity (J)
H	Battery state of health (SOH)
$R_{\text{int}}(T, s)$	Internal battery resistance (Ω)
$x_{\text{net}}(t)$	Network activity intensity (Mbps)
$z(t)$	Network tail energy state
TTE	Time-to-empty
MSE	Mean squared error of SOC prediction

4 Continuous-Time SOC Model

4.1 Simplest Continuous Battery Drain Model

Battery drain refers to the continuous depletion of stored energy in a lithium-ion smartphone battery driven by the device's power consumption. In its simplest form, the battery level decreases smoothly over time due to a combination of factors: screen usage, processor workload, network connectivity and data transmission, GPS activity, and background system processes. This forms our starting point

that the battery is treated as an energy reservoir that is steadily depleted by the device's total power consumption, serving as the baseline model. The governing equation can be written as:

$$\frac{ds}{dt} = -\kappa P_{\text{tot}}(t) \quad \kappa = \frac{100}{E_{\text{max}}} \quad (1)$$

In this formulation, $s(t)$, the state of charge (SOC), expressed as a percentage of the battery's maximum available energy E_{max} , and $P_{\text{tot}}(t)$ represents the device's total instantaneous power consumption in watts (W). The negative sign reflects the fact that battery charge decreases as energy is consumed. The coefficient $\kappa = 100/E_{\text{max}}$ serves as a scaling factor that converts power consumption (J/s) into a rate of change in SOC per unit time. Consequently, the SOC dynamics are governed by the device's total power consumption.

Empirical studies suggests that the total power consumption can be decomposed into the sum of contributions from major hardware subsystems including backlight, GSM, CPU, RAM, Network interfaces (GSM, Wifi), GPS, and other components [1]. This inspire us to decompose $P_{\text{tot}}(t)$ into interpretable components corresponding to screen usage, processor load, network activity, GPS, and background tasks to refine the model.

$$P_{\text{tot}}(t) = P_{\text{base}} + P_{\text{scr}}(t) + P_{\text{cpu}}(t) + P_{\text{net}}(t) + P_{\text{gps}}(t) + P_{\text{bg}}(t) \quad (2)$$

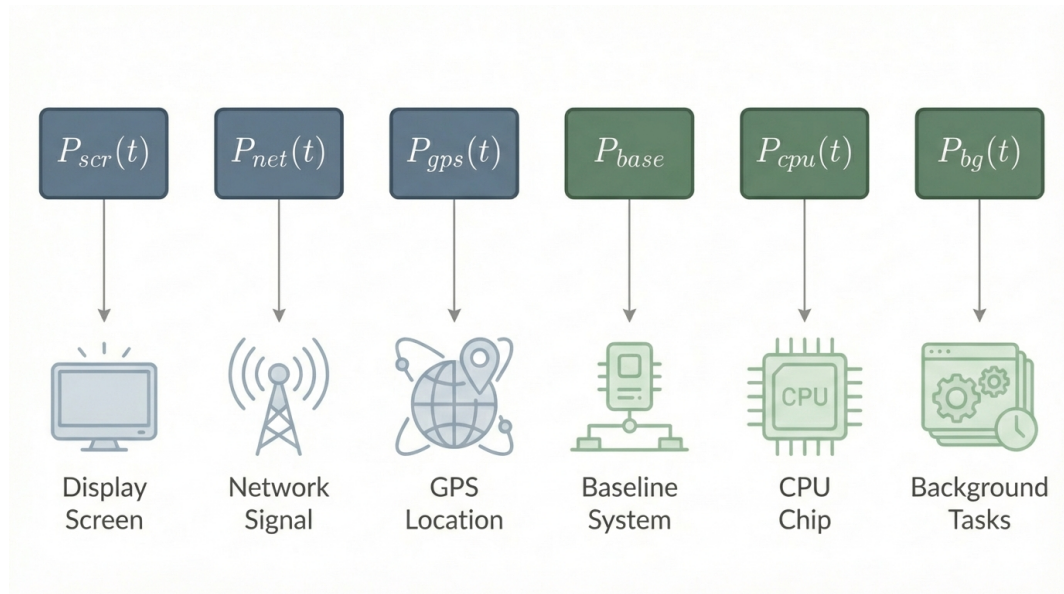


Figure 2: Simplest Continuous Battery Drain Model

4.2 Improved SOC Model A Coupled Dynamical System

4.2.1 A Generalized Dynamic Representation

While the baseline model captures battery drain as a direct consequence of total power consumption, real-world smartphone behavior is influenced by additional factors that cannot be fully represented by

instantaneous power demand alone. In particular, usage patterns, system-level controls, and internal physical states introduce nonlinearities, delays, and feedback effects in the SOC evolution.

To accommodate these effects within a unified framework, we represent the SOC dynamics in a generalized continuous-time form:

$$\frac{ds}{dt} = -F(s(t), u(t), z(t)) \quad (3)$$

where $s(t)$ denotes the state of charge (SOC) expressed as a percentage, $u(t)$ represents a vector of external inputs describing user activity and system workload (e.g., screen usage, brightness level, CPU load, network activity), and $z(t) \geq 0$ denotes internal state variables representing latent system dynamics, including network tail states, thermal states, and internal loss mechanisms.

The detailed function F is:

$$\frac{ds}{dt} = -\frac{100}{E_{\text{cap},J}(T, H)} (P_{\text{sys}}(t) + P_{\text{loss}}(t)) \quad (4)$$

where $P_{\text{sys}}(t)$ is the total system power consumption due to device operation (screen, CPU, network, etc.), $P_{\text{loss}}(t)$ is the internal power loss caused by internal resistance, and $E_{\text{cap},J}(T, H)$ is the effective usable battery energy capacity (in joules). The factor 100 converts energy depletion into a percentage change of SOC, and the negative sign indicates that SOC decreases as energy is consumed.

Energy capacity is a function of temperature T (in Kelvin) and state of health (SOH) H :

$$E_{\text{cap},J}(T, H) = 3600 E_{\text{nom},\text{Wh}} \eta_{\text{temp}}(T) H \quad (5)$$

where $E_{\text{nom},\text{Wh}}$ denotes the nominal battery energy capacity, $\eta_{\text{temp}}(T)$ characterizes the effect of temperature on usable capacity, and H represents capacity loss due to battery aging. The SOC evolution depends on both power consumption and usable capacity.

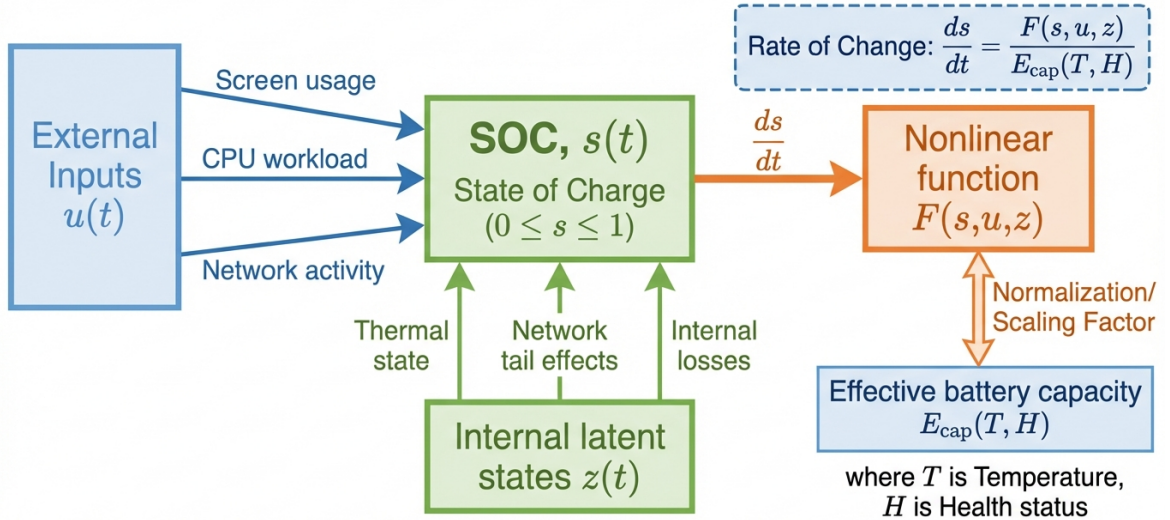


Figure 3: Continuous-Time SOC Model

The state of health (SOH), denoted by $H \in (0, 1]$, is introduced as a scalar measure of battery [2] and is defined as the ratio between the current usable capacity and the initial nominal capacity,

$$H = \frac{C_a}{C_0} \quad (6)$$

and is assumed to depend only on the accumulated charge–discharge cycles.

A fresh battery corresponds to $H = 1$; an aged battery with accumulated degradation corresponds to $H < 1$. As the battery ages, its effective capacity decreases, means that even at the same SOC level, an aged battery contains less usable energy than a healthy one. Consequently, SOC appears to deplete faster for batteries with lower SOH due to the reduced total available capacity.

However, as mentioned in Assumption 1, we assume $H = 1$ because the model focuses on short-term SOC dynamics following a single charging event, over which battery aging is negligible and SOH can be treated as constant.

In the following sections, we develop the governing equations of the model by considering plausible phenomenons, employing fundamental physical principles and incorporating insights from literature.

4.2.2 Temperature's Effect on Effective Battery Capacity, $\eta_{\text{temp}}(T)$

Empirically, batteries operating at the same state of charge (SOC) can exhibit substantially different runtime under different temperature conditions, indicating that treating battery capacity as temperature-independent is insufficient for accurately modeling SOC evolution.

Previous studies show that lithium-ion battery discharge capacity depends nonlinearly and non-monotonically on temperature: usable capacity increases with temperature, peaks within an optimal range, and degrades at both low and excessively high temperatures [3]. This behavior has been shown to be well approximated by a cubic polynomial over the typical operating temperature range, with optimal performance occurring around 20–50 °C [3].

Motivated by these findings, we model the temperature dependence of usable battery capacity using a cubic polynomial and introduce a temperature-dependent capacity correction factor $\eta_{\text{temp}}(T)$ to modify the effective energy capacity in the SOC dynamics. Specifically, the usable capacity is given by

$$C(T) = 36.713 (T - 273.15)^3 - 826.14 (T - 273.15)^2 + 5988.6 (T - 273.15) - 4204.4 \quad (7)$$

where T is the battery temperature in Kelvin. Normalizing at the reference temperature $T_{\text{ref}} = 298.15$ K,

$$\eta_{\text{temp}}(T) = \frac{C(T)}{C(T_{\text{ref}})} \quad (8)$$

ensures $\eta_{\text{temp}}(T_{\text{ref}}) = 1$, while reduced capacity is reflected by $\eta_{\text{temp}}(T) < 1$ under non-optimal thermal conditions.

4.2.3 Thermal Throttling Feedback Mechanism, $\phi(T)$

In real-world usage scenarios, smartphone power consumption does not increase linearly with usage intensity. Under sustained high-load operation, the internal temperature of the device gradually rises, triggering system-level thermal management mechanisms such as CPU frequency scaling, dynamic voltage and frequency scaling (DVFS), and display brightness limiting [4]. These mechanisms actively suppress power consumption under overheating conditions, thereby slowing further temperature increase. This interaction gives rise to a feedback mechanism in which power consumption leads to temperature rise, while elevated temperature in turn constrains power consumption. Ignoring this mechanism would prevent the model from capturing commonly observed nonlinear battery behavior, where battery drain is rapid during the initial phase of high load and gradually slows thereafter.

Existing studies typically adopt threshold-based control strategies, whereby system frequency or resource usage is reduced once the chip temperature approaches a predefined thermal limit, in order to prevent hardware damage [5]. However, such mechanisms are often implemented at the operating system or hardware level via discrete state switching, which is not directly compatible with a continuous-time SOC dynamical framework.

Motivated by the literature, and to account for both the observed nonlinear battery drain under sustained load and the requirement of a continuous-time model, we introduce a temperature-dependent throttling factor $\phi(T)$ to approximate the suppressive effect of thermal constraints on power components.

$$\phi(T) = \frac{1}{1 + e^{\lambda(T - T_{\text{thresh}})}} \quad (9)$$

As shown in equation (9), the throttling factor is modeled using a logistic function, enabling a smooth transition in power suppression as temperature varies. When the temperature remains below the threshold, $\phi(T) \approx 1$, indicating negligible thermal restriction. As the temperature approaches or exceeds the threshold, $\phi(T)$ decreases gradually, representing increasing suppression of active power consumption through mechanisms such as frequency scaling and brightness limiting. By embedding this throttling factor into power consumption model, the resulting SOC dynamics are able to capture nonlinear battery behavior arising from the coupling between temperature and power consumption.

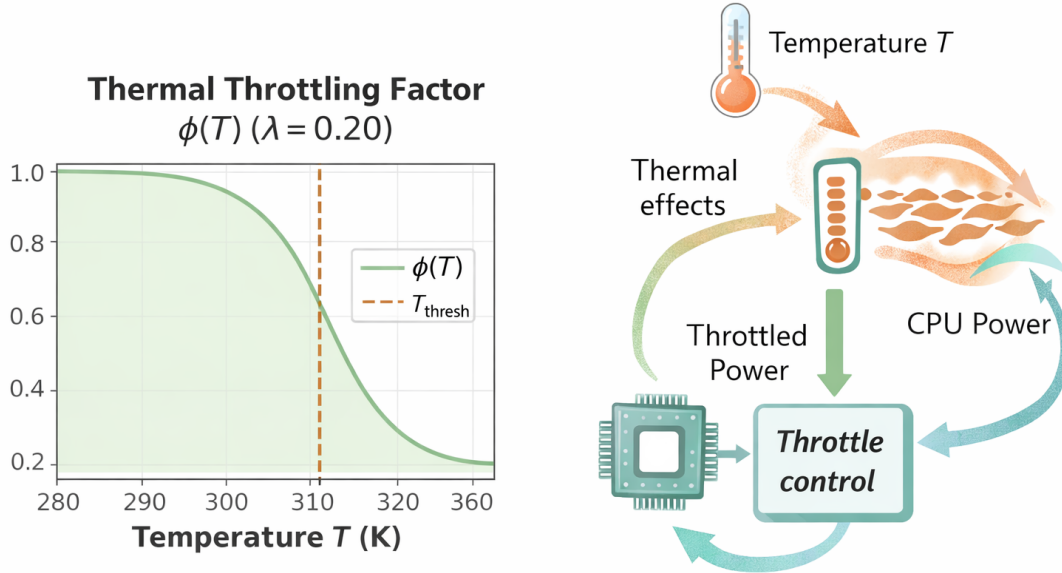


Figure 4: Thermal Throttling Feedback Mechanism

4.2.4 Network Power Consumption with Tail Energy, $P_{\text{net}}(t)$

Network power consumption is strongly influenced by tail energy effects, whereby wireless interfaces remain in elevated power states for a period after data transmission ends, especially for cellular networks [6, 7]. This behavior arises from radio resource control (RRC) state machines and inactivity timers, and cannot be captured by instantaneous traffic demand alone.

To model this effect, we introduce a network tail state $z(t)$ governed by a first-order differential equation,

$$\frac{dz}{dt} = \frac{1}{\tau_{\text{net}}^{(\text{type})}} (x_{\text{net}}(t) - z(t)) \quad (10)$$

where $x_{\text{net}}(t)$ denotes instantaneous network activity intensity and $\tau_{\text{net}}^{(\text{type})}$ is a time constant characterizing the tail duration for different network interfaces. Cellular networks exhibit larger $\tau_{\text{net}}^{(\text{type})}$ than WiFi, reflecting their more persistent high-power tail behavior.

The resulting network power consumption is modeled as

$$P_{\text{net}}(t) = \frac{1}{S_{\text{signal}}(t)} \left(P_{\text{idle}}^{(\text{type})} + k_{\text{net}}^{(\text{type})} z(t) \right) \quad (11)$$

where $P_{\text{idle}}^{(\text{type})}$ represents the baseline idle power of the active network interface and $k_{\text{net}}^{(\text{type})}$ scales the contribution of tail activity to power consumption. Differences between WiFi and cellular networks are captured through these parameters, with cellular interfaces exhibiting higher idle power and stronger tail effects.

To account for real-world wireless conditions, a signal strength correction factor $S_{\text{signal}}(t)$ is introduced. Poor signal quality ($S_{\text{signal}}(t) < 1$) increases transmission power requirements and therefore amplifies network energy consumption, without explicitly modeling lower-layer communication dynamics.

This formulation captures the persistence of network energy consumption beyond active data transmission and provides a compact, scenario-aware representation of network power dynamics.

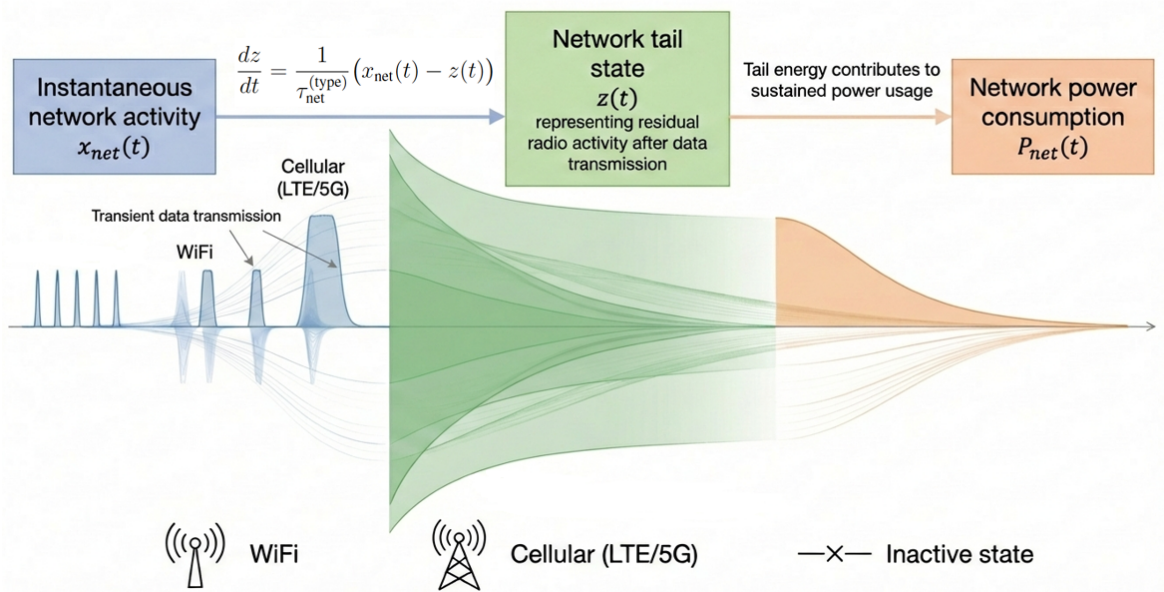


Figure 5: Dynamic modeling of network power consumption with tail energy effects. Instantaneous network activity excites a latent tail state governed by a first-order differential equation, whose decay rate depends on the network type. The residual tail state contributes to sustained network power consumption even after data transmission ends.

4.2.5 System Power Decomposition, $P_{\text{sys}}(t)$

With thermal throttling feedback incorporated, total system power can no longer be expressed as a purely additive function of component loads. In particular, screen and CPU power are regulated by a temperature-dependent throttling factor $\phi(T)$, reflecting system-level performance control under thermal constraints. The resulting system power is decomposed as

$$P_{\text{sys}}(t) = P_{\text{base}} + \phi(T)(P_{\text{scr}}(u_{\text{on}}, b) + P_{\text{cpu}}(u_{\text{cpu}})) + P_{\text{net}}(z, S_{\text{signal}}) \quad (12)$$

Screen power is modeled as an activation-weighted function of brightness, abstracting pixel-level details while remaining sensitive to user-controlled settings,

$$P_{\text{scr}}(t) = u_{\text{on}}(t)(\alpha_0 + \alpha_1 b(t)) \quad (13)$$

where $u_{\text{on}}(t)$ indicates whether the screen is active and $b(t)$ denotes the brightness level.

CPU power is represented using a reduced-form nonlinear utilization model that captures the superlinear relationship between computational load and power under dynamic voltage and frequency scaling (DVFS),

$$P_{\text{cpu}}(t) = k_{\text{cpu}}(3 u_{\text{cpu}}(t))^3 \quad (14)$$

where $u_{\text{cpu}}(t)$ is the normalized CPU utilization and hardware-dependent effects are absorbed into the coefficient k_{cpu} .

Network power consumption $P_{\text{net}}(t)$ is modeled separately to account for tail energy dynamics, signal quality, and network type-dependent behavior, as shown in 4.2.4.

4.2.6 Loss via Internal Resistance, $P_{\text{loss}}(t)$

Empirical observations indicate that battery depletion accelerates under low temperature, high load, and extreme SOC conditions, which cannot be captured by a linear power-to-SOC relationship. This behavior arises from irreversible internal losses associated with battery internal resistance, through which a portion of the extracted energy is dissipated as heat [2].

We model the internal resistance as a coupled function of temperature and SOC. The temperature dependence is described using a reference-normalized Arrhenius relation,

$$R_{\text{int}}^{\text{temp}}(T) = R_{\text{ref}} \exp \left[\frac{E_a}{k_B} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}} \right) \right] \quad (15)$$

where R_{ref} is the resistance at reference temperature T_{ref} , E_a denotes the effective activation energy, and k_B is the Boltzmann constant. This formulation captures the thermally activated nature of ion transport and charge-transfer processes while enabling stable calibration relative to a known operating condition.

To account for SOC-dependent resistance growth at extreme charge levels, we introduce a nonlinear correction factor,

$$R_{\text{int}}(T, s) = R_{\text{int}}^{\text{temp}}(T) \left[1 + c \left(\frac{s_{\text{ref}}}{s + \varepsilon} - 1 \right) \right] \quad (16)$$

where s is the SOC, s_{ref} is a reference SOC, c controls sensitivity to SOC variation, and ε prevents singularity as $s \rightarrow 0$. This correction leaves resistance unchanged near s_{ref} and induces a sharp increase at low SOC, consistent with empirical discharge behavior.

For tractability, the discharge current is approximated using the nominal voltage,

$$I(t) \approx \frac{P_{\text{sys}}(t)}{V_{\text{nom}}} \quad (17)$$

yielding the internal-resistance power loss

$$P_{\text{loss}}(t) = \frac{P_{\text{sys}}(t)^2}{V_{\text{nom}}^2} R_{\text{int}}(T, s) \quad (18)$$

This formulation introduces a nonlinear feedback pathway through which temperature, SOC, and power demand jointly accelerate battery depletion under adverse operating conditions.

4.2.7 Thermodynamics, $P_{\text{heat}}(t)$

In practice, smartphone temperature evolves with usage intensity: power draw during activities such as video calling or gaming generates internal heat, and reduced usage allows the device to cool toward ambient conditions.

Starting from energy conservation, this behavior is described by

$$C \frac{dT}{dt} = P_{\text{heat}} - hA(T - T_{\text{env}}) \quad (19)$$

where the temperature dynamics result from the balance between internal heat generation and heat dissipation to the environment. The formula models the rate of stored thermal energy with the employment of Newton's law of cooling for heat dissipation.

Based on the above thermodynamic model, we model temperature as a dynamic state variable coupled with power consumption and battery behavior. Power consumption generates internal heat that governs temperature evolution, while temperature in turn affects effective battery capacity and the discharge process, forming the thermodynamic submodule of our improved model.

In our model, we write

$$\frac{dT}{dt} = \frac{1}{C_{\text{eq}}} \left(P_{\text{heat}}(t) - hA(T(t) - T_{\text{env}}(t)) \right) \quad (20)$$

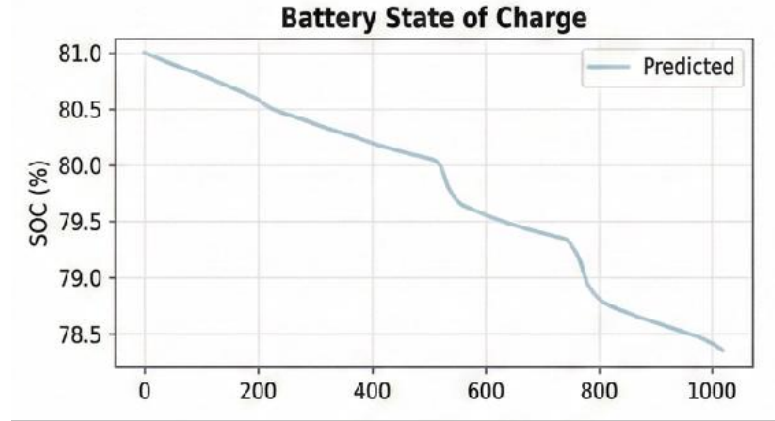
where C_{eq} calculated in J/K is the equivalent heat capacity that combine the thermal mass and specific heat into a single equivalent heat capacity, hA (W/K) is the equivalent heat transfer coefficient; we approximate the internal heat generation $P_{\text{heat}}(t)$ as

$$P_{\text{heat}}(t) \approx P_{\text{sys}}(t) + P_{\text{loss}}(t) \quad (21)$$

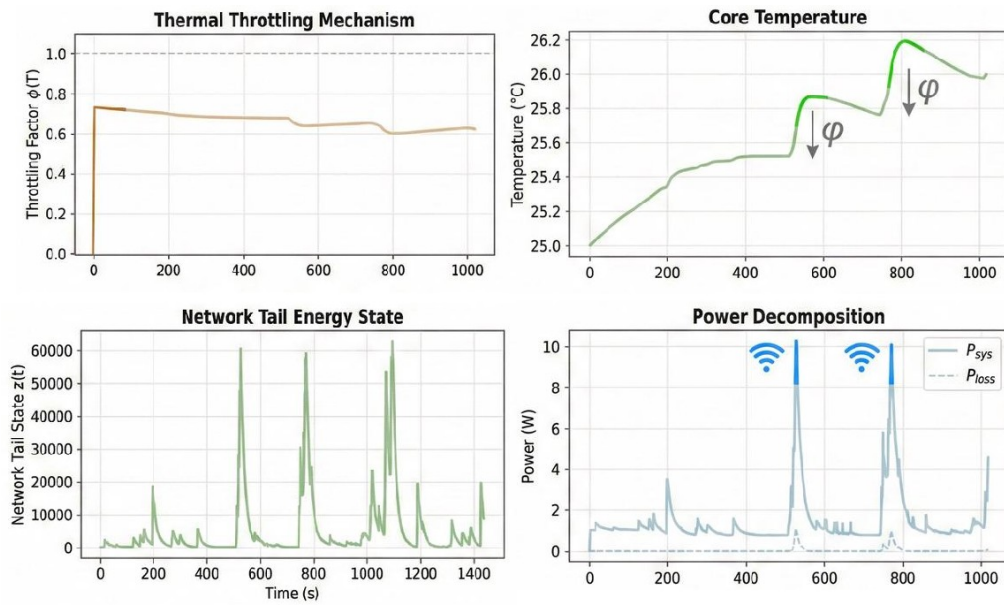
with the two terms $P_{\text{sys}}(t)$ and $P_{\text{loss}}(t)$ defined in 4.2.5 and 4.2.6 respectively.

4.2.8 Results

Our data is sourced from Sherlock [8]. As shown by the Network Tail Energy State in Figure 6, prominent peaks are observed approximately within the time intervals of 400–600 s and 700–900 s. These peaks induce corresponding increases in P_{sys} over the same periods, which in turn drive the temperature to reach local maxima. Owing to the negative feedback inherent in the thermal feedback mechanism, a rapid rise in temperature leads to a gradual decrease in ϕ . This reduction constrains the output of P_{sys} , causing the temperature to subsequently decline. Meanwhile, during intervals of elevated power consumption and rising temperature, the SOC decreases at an accelerated rate, resulting in a nonlinear variation in the SOC–t curve.



(a) SOC



(b) Temperature and Power Decomposition

Figure 6: Illustrative applications of the proposed battery and power consumption model.

5 Time to Empty Estimation

5.1 The Data

5.1.1 Data Collection

The dataset used in this study is obtained from an openly accessible dataset: “Tour de France” which is published on ScienceDirect [9]. The dataset covers five representative smartphone usage scenarios commonly observed in daily operation, as summarized in Table 1.

These scenarios span a wide range of screen usage, CPU load, and network activity levels, allowing us to evaluate battery discharge behavior under diverse and realistic operating conditions.

Table 1: Representative smartphone usage scenarios in the dataset

Scenario	Key characteristics
Music playback	Low screen usage, low CPU, minimal network
Social media browsing	Active screen, moderate CPU, background GPS/network
Online video streaming (WiFi)	Active screen, sustained network, moderate CPU
Local video playback	Active screen, minimal network, moderate decoding
Mobile gaming	Active screen, high CPU/GPU, intermittent network

5.1.2 Data Processing

In the data processing phase, we introduce a network-related variable $x_{\text{net}}(t)$ (Mbps) to characterize the network transmission intensity induced by user interactions. Its value varies across usage scenarios, reflecting differences in network access patterns, with music playback exhibiting low intermittent traffic, while video streaming and gaming produce higher and more sustained loads.

Social Media Scenario In social media browsing (e.g., Facebook), network activity mainly arises from intermittent page loading and periodic content refreshing. This sporadic yet structured traffic pattern is modeled as

$$x_{\text{net}}(t) = \max \left\{ x_0 + A \sin \left(\frac{2\pi t}{\tau_{\text{page}}} \right), 0 \right\} \quad (22)$$

where $\tau_{\text{page}} = 60$ s denotes the page refresh period, A controls the fluctuation amplitude, and the truncation enforces non-negativity of network traffic intensity.

Music scenario In music playback scenario, the same sinusoidal model is used because network activity is mainly driven by lightweight and periodic background synchronization, consistent with the buffering and keep-alive mechanisms commonly employed in mobile operating systems.

Online Video Streaming Scenario In online video streaming (e.g., 1080p playback), network transmission follows a chunk-based mechanism, leading to periodically occurring high-load bursts separated by low-activity intervals. The instantaneous network activity is modeled as

$$x_{\text{net}}(t) = \begin{cases} \min(x_{\text{peak}}, \max(x_{\text{base}}, 0.6 x_{\text{peak}} Y_t)) & t \bmod \tau_{\text{chunk}} < \tau_{\text{load}} \\ \varepsilon_t & \text{otherwise} \end{cases} \quad (23)$$

Here, x_{base} and x_{peak} denote the baseline and peak network load during active downloading, τ_{chunk} is the chunk period, and τ_{load} is the corresponding download duration. The burst amplification factor Y_t follows a heavy-tailed Pareto distribution,

$$Y_t \sim \text{Pareto}(y_{\min}, \alpha) \quad y_{\min} = 1, \alpha = 1.5 \quad (24)$$

where the lower bound $y_{\min} = 1$ corresponds to a unit-scale Pareto distribution, ensuring positivity and finite expectation for $\alpha > 1$. This choice prevents the burst multiplier from becoming arbitrarily small during active downloading, while the heavy-tailed property allows for rare but extreme throughput spikes consistent with empirical streaming behavior. The term ε_t represents low-level background network activity outside downloading windows.

Gaming Scenario In gaming scenarios, network traffic consists of persistently transmitted high-frequency packets with randomly triggered short-lived synchronization bursts. Compared with video streaming, gameplay exhibits minimal low-activity intervals and a smaller separation between baseline and peak network load.

Building on the framework of equation (23), gaming traffic is modeled as a bounded stochastic process with an elevated baseline load to reflect continuous heartbeat and control communication. Burst events are modeled as randomly triggered and heavy-tailed to capture rare but intense spikes, while Gaussian perturbations are introduced during non-burst periods to represent continuous micro-fluctuations, yielding a more temporally uniform and stable network traffic pattern.

5.2 Time-to-Empty (TTE) Under Different Initial SOC and Usage Scenarios

For a given usage scenario and initial SOC level, the time-to-empty (TTE) is defined as the time required for the SOC trajectory to reach zero, i.e., the first time instant at which the solution of the SOC differential equation satisfies $\text{SOC}(t) = 0$. TTE serves as an aggregate performance metric summarizing the cumulative impact of power consumption under a specific usage pattern.

We evaluate TTE for three evenly spaced initial SOC values,

$$\text{SOC}_0 \in \{33\%, 66\%, 100\%\} \quad (25)$$

representing low, medium, and high initial charge states. For each SOC_0 , the SOC dynamics from Task 1 are simulated under five representative usage scenarios (music playback, social media browsing, online video streaming, local video playback, and mobile gaming), and TTE is obtained by integrating the SOC dynamics forward in time until depletion.

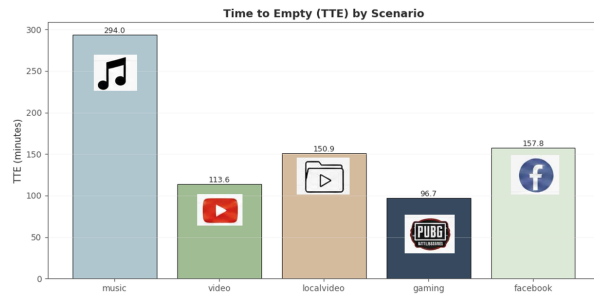
The results are analyzed by jointly examining TTE values, SOC depletion trajectories, and temperature responses, revealing how differences in workload intensity translate into distinct depletion rates and time-to-empty outcomes across usage scenarios.

Across all three initial SOC levels, the model consistently explains differences in TTE through scenario-dependent power consumption and thermal response. Regardless of the initial SOC, the relative ordering of TTE remains stable: music playback > social media browsing > local video playback > online video streaming > mobile gaming.

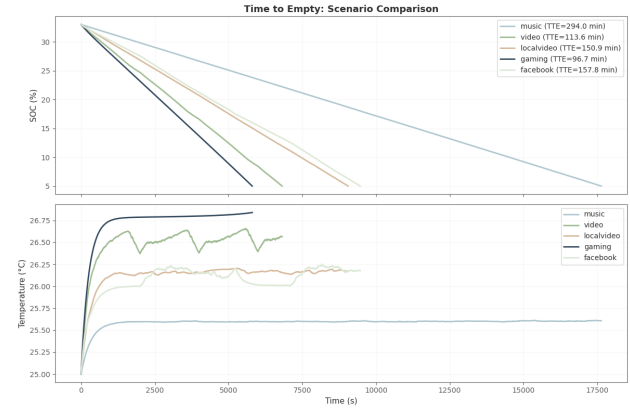
From Figures 7b, 7d, and 7f, SOC depletion is observed to be approximately linear across scenarios. Mobile gaming exhibits the steepest decline, indicating the highest sustained power consumption, whereas music playback shows the slowest depletion.

Temperature trajectories display similar qualitative patterns across initial SOC levels. High-intensity scenarios, such as mobile gaming and online video streaming, induce faster temperature rise and larger thermal fluctuations, while low-power scenarios remain close to thermal steady state. This indicates that thermal behavior is primarily driven by instantaneous power demand rather than the available charge level.

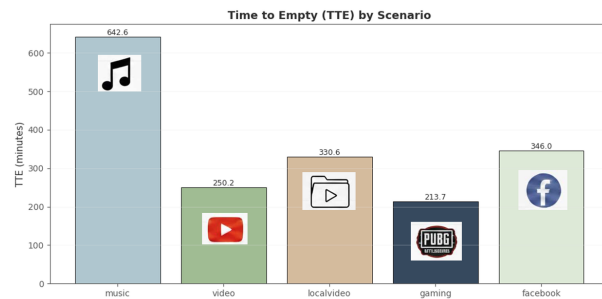
Increasing the initial SOC scales TTE approximately proportionally, while the relative ordering of scenarios and the shapes of SOC and temperature trajectories remain unchanged, suggesting that initial SOC affects operating duration but not the underlying consumption dynamics.



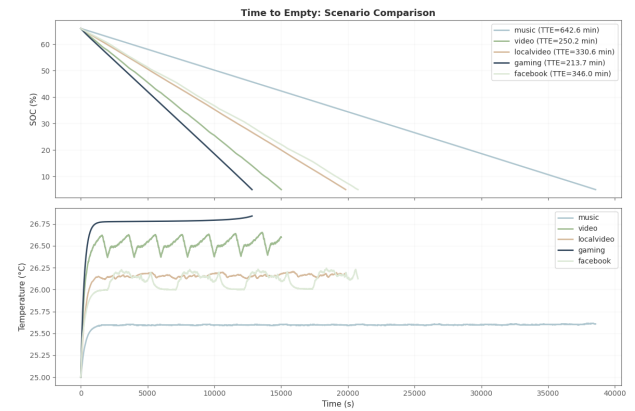
(a) TTE comparison (33% SOC).



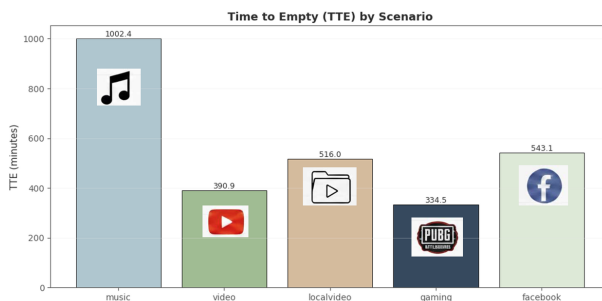
(b) SOC and temperature trajectories (33% SOC).



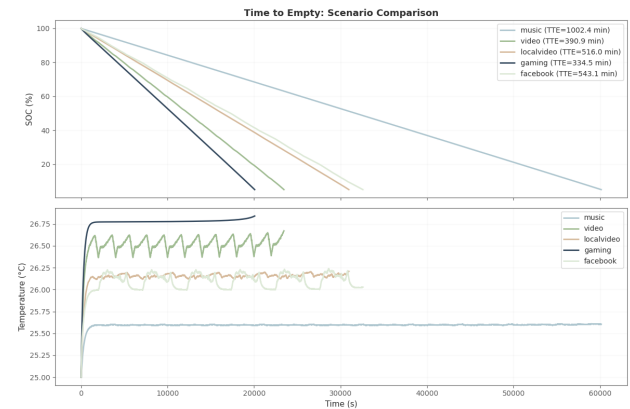
(c) TTE comparison (66% SOC).



(d) SOC and temperature trajectories (66% SOC).



(e) TTE comparison (100% SOC).



(f) SOC and temperature trajectories (100% SOC).

Figure 7: Time-to-empty (TTE), SOC depletion, and temperature evolution under different initial SOC levels.

5.3 Model Performance Evaluation via MSE

To evaluate how well the proposed SOC model aligns with observed (or plausible) real-world behavior, we compare model-predicted SOC trajectories with available SOC observations under two representative scenarios: music playback and online video streaming, both evaluated at an initial SOC of 33%. These two scenarios exhibit markedly different load characteristics and therefore provide a meaningful test of the model's ability to distinguish usage-dependent depletion dynamics.

Because the observed SOC traces originate from different absolute levels (approximately 90% for music and 40% for video), while the proposed model assumes a smoothed SOC evolution with extrapolated initial conditions (Assumption 2), the comparison is restricted to the overlapping time intervals for which observed SOC data are available.

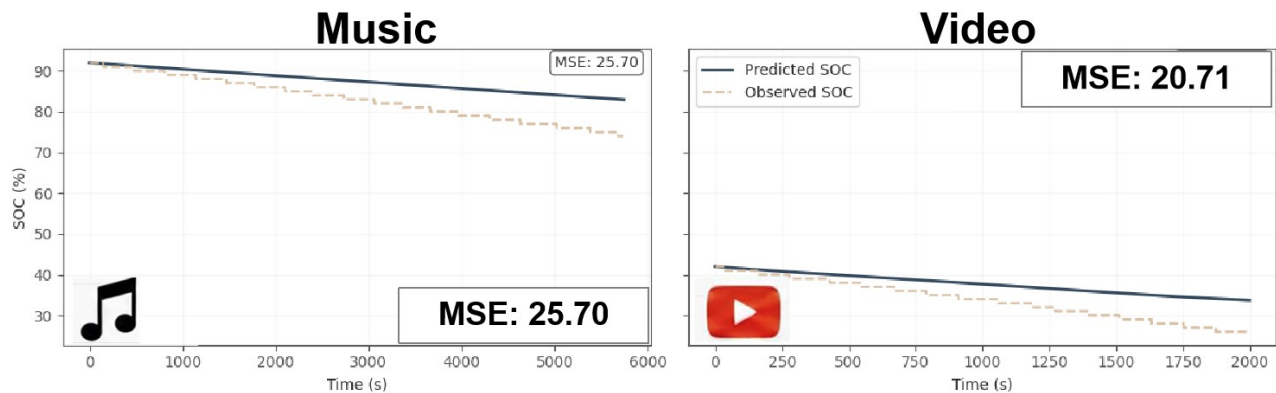


Figure 8: Model evaluation via MSE between predicted and observed SOC trajectories.

Model accuracy is quantified using the mean squared error (MSE) between predicted and observed SOC trajectories. The resulting MSE values are 25.70 for music playback and 20.71 for online video streaming, corresponding to root mean squared errors (RMSE) of approximately 4.5–5.1% on a 0–100% SOC scale, indicating moderate agreement.

Importantly, this level of error is consistent with the intended scope of the model. As a framework that emphasizes usage-driven power demand, thermal feedback, and capacity effects. The model is not calibrated to device-specific battery lifetime data; instead, it prioritizes structural trend fidelity to demonstrate that relative TTE differences arise from model mechanisms rather than parameter fitting.

Model strengths

Despite its simplicity, the model captures overall SOC depletion trends and relative slopes across scenarios, indicating that the dominant mechanisms—scenario-dependent power demand, thermal feedback, and effective capacity effects—are correctly integrated and sufficient to explain first-order differences in battery behavior.

Model limitations

Discrepancies in absolute SOC values mainly arise from uncalibrated parameters and manually tuned coefficients, including scenario-level power scaling, simplified thermal response, and coarse capacity and health adjustments. Addressing these limitations would require device-specific measurements and data-driven parameter estimation, which are beyond the scope of this study.

5.4 Results

5.4.1 Identify the Drivers of Rapid Battery Drain

Power consumption is the direct driver of SOC depletion. Consequently, the relative distribution of power among system components reflects the importance of different drivers of battery drain.

Based on this principle, we simulate a smartphone operating under different usage scenarios and visualize the resulting power contributions using pie charts. Note that the initial SOC is fixed at 100 % in this analysis, since Section 5.2 shows that initial SOC has negligible impact on the relative TTE behavior across scenarios. The results are shown in figure 9.

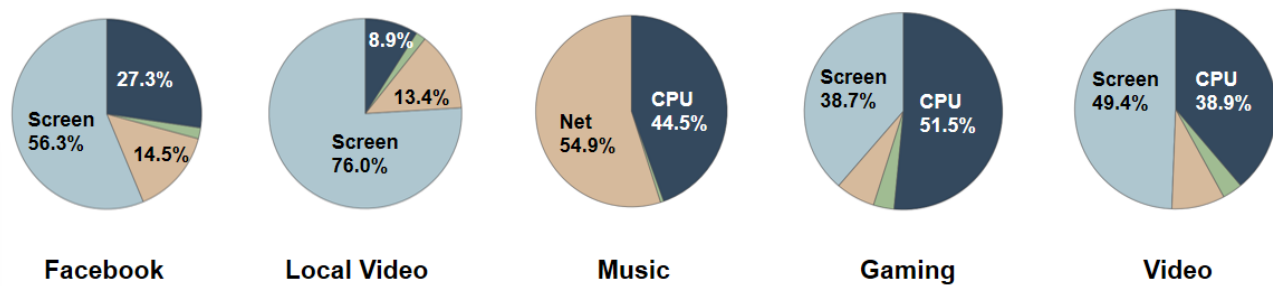


Figure 9: Power contribution pie chart for each scenario

Music playback The dominant contributor to power consumption is the network, accounting for 54.9% of total power consumption. In this scenario, the screen is largely inactive and CPU load is low, while periodic data transmission and network tail energy dominate overall power usage.

Facebook browsing The dominant contributor to power consumption is the screen, accounting for 56.3% of total power consumption. This is because social media browsing involves sustained screen-on time with moderate brightness, while CPU and network activities remain relatively lightweight and intermittent.

Local video playback The dominant contributor to power consumption is the screen, accounting for 76.0% of total power consumption. Local video playback requires continuous display activity at elevated brightness, while network usage is minimal and decoding-related CPU load remains moderate.

Video playback Power consumption is primarily driven by a combination of the screen (49.4%) and CPU (38.9%). Online video streaming requires sustained display activity together with continuous video decoding and buffering, leading to simultaneous contributions from screen and processor power.

Mobile gaming The dominant contributor to power consumption is the CPU, accounting for 51.5% of total power consumption. Mobile gaming induces sustained high computational load due to real-time graphics rendering and frequent logic updates, which directly increases dynamic CPU power in the model.

5.4.2 Activities Producing the Greatest and Least Battery Life Reduction

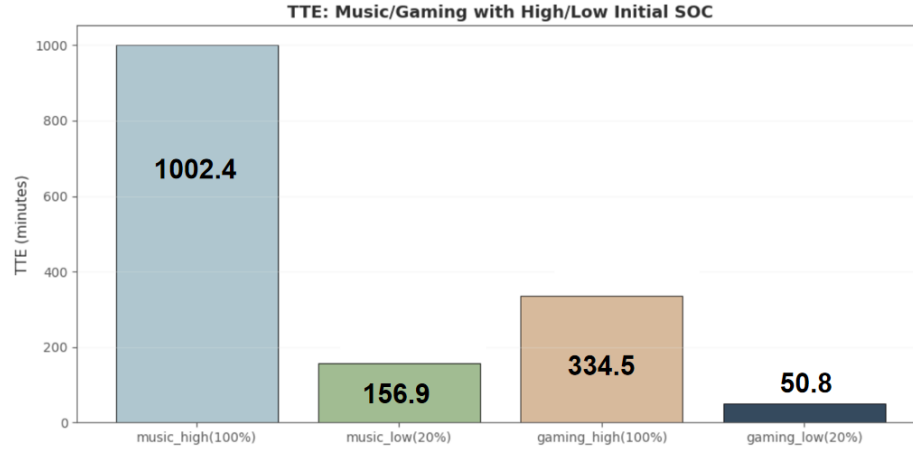


Figure 10: TTE comparison for music playback and mobile gaming under high (100%) and low (20%) initial SOC conditions

Mobile gaming produces the greatest reduction in battery life across both initial SOC levels. As shown above, the predicted TTE under gaming is substantially shorter than under music playback.

This behavior is explained by the model’s power–thermal coupling. Mobile gaming sustains high CPU utilization and active graphics rendering, leading to elevated system power consumption. According to the thermal dynamics incorporated in the model, higher power input increases device temperature, which in turn raises internal resistance and accelerates effective energy depletion. Prolonged operation at elevated temperatures is also associated with long-term battery degradation (i.e., SOH aging), thus making gaming as the dominant driver of rapid battery drain in the model.

By contrast, music playback changes the model outcomes surprisingly little, especially in terms of relative TTE behavior. Although reducing the initial SOC from 100% to 20% significantly shortens absolute TTE, the qualitative behavior under music playback remains stable, with consistently shallow SOC decline and low thermal stress.

This is because music playback operates under relatively low and steady power consumption, dominated by lightweight CPU usage and limited screen activity. As a result, device temperature remains close to ambient levels, making thermal feedback play a minimal role. The model therefore predicts similar discharge dynamics regardless of thermal or aging-related effects, making music playback relatively insensitive to changes in operating conditions.

6 Sensitivity Analysis

6.1 Relaxing the Constant SOH Assumption

In Tasks 1 and 2, the model focuses on short-term, single-cycle discharge dynamics, under which SOH evolves much more slowly than SOC and temperature and is therefore treated as a constant.

To examine the robustness of this assumption, we relax the constant-SOH setting and explicitly investigate how variations in SOH affect SOC evolution. We consider the same usage scenario (social

media usage) and the same initial SOC level (e.g., 60%), while varying SOH across representative levels.

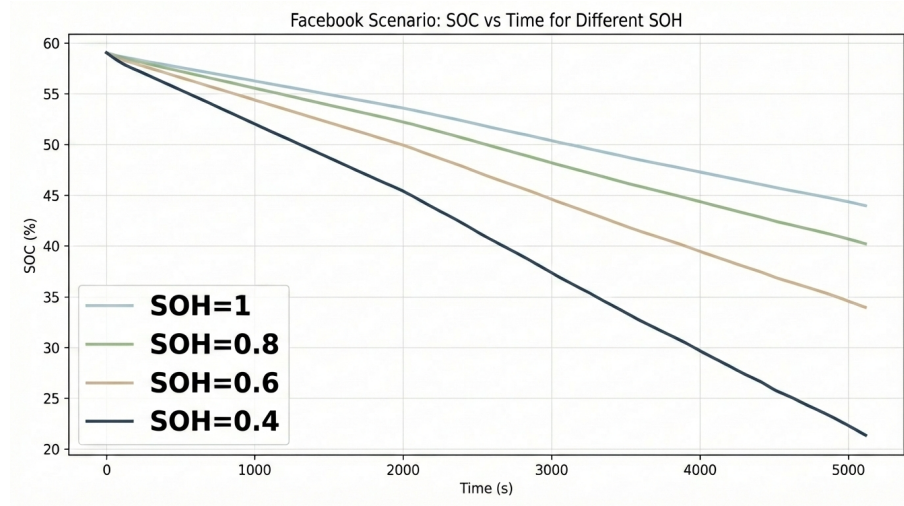


Figure 11: SOC evolution over time under the Facebook usage scenario for different battery SOH levels.

The results (Figure 11) show that lower SOH leads to a faster SOC decline. As SOH decreases from 1.0 to 0.4, the SOC trajectories exhibit steeper slopes under the same usage pattern, reflecting reduced effective capacity. While SOH does not alter the qualitative shape of the SOC trajectory, it significantly affects the absolute depletion rate, indicating that SOH primarily acts as a scaling factor on available capacity rather than a driver of short-term dynamics.

6.2 Adjusting Parameter in Thermal Throttling Factor ϕ

Thermal throttling is modeled through a temperature-dependent suppression factor $\phi(T)$, parameterized by a threshold temperature T_{thresh} and a slope parameter λ . Since these parameters are device-dependent and not directly observable, a sensitivity analysis is performed by varying T_{thresh} and λ while holding other factors fixed.

The results (Figure 12) show that higher T_{thresh} and larger λ lead to increased SOC decline rates by allowing higher power consumption or sharper throttling transitions. Across the explored parameter range, the SOC decline rate varies smoothly without regime shifts, indicating that thermal throttling scales the magnitude of battery drain rather than altering the qualitative discharge behavior. Overall, the model's short-term SOC dynamics remain structurally robust to variations in T_{thresh} and λ .

6.3 Fluctuation in Usage Patterns

In Task 2, usage scenarios are assumed to be stationary, whereas real-world smartphone usage typically consists of multiple activities concatenated over time. To capture this variability, we construct a composite usage process combining music playback, video streaming, and mobile gaming, with each scenario lasting a random duration between 10 and 40 minutes and transitions governed by a Markov process.

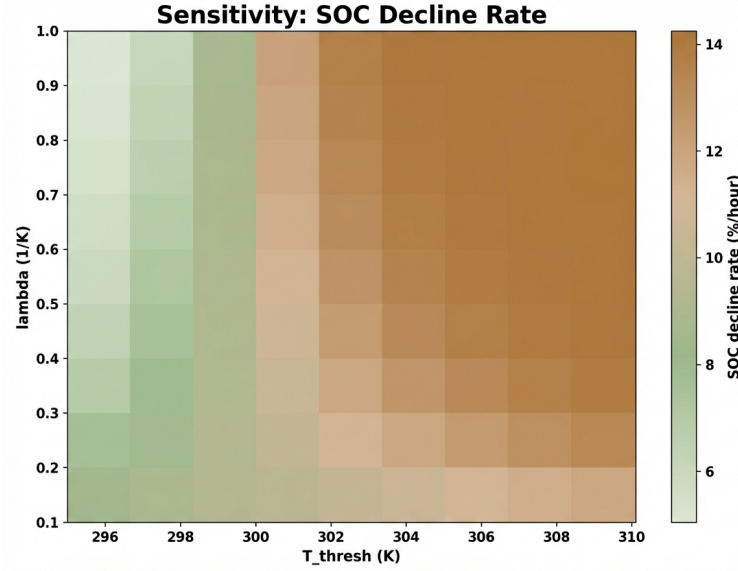


Figure 12: Sensitivity of the SOC decline rate to the thermal throttling parameters T_{thresh} and λ . The heatmap shows the predicted SOC decline rate (percent per hour) under a fixed usage scenario as the thermal throttling onset temperature T_{thresh} and the throttling sharpness parameter λ vary.

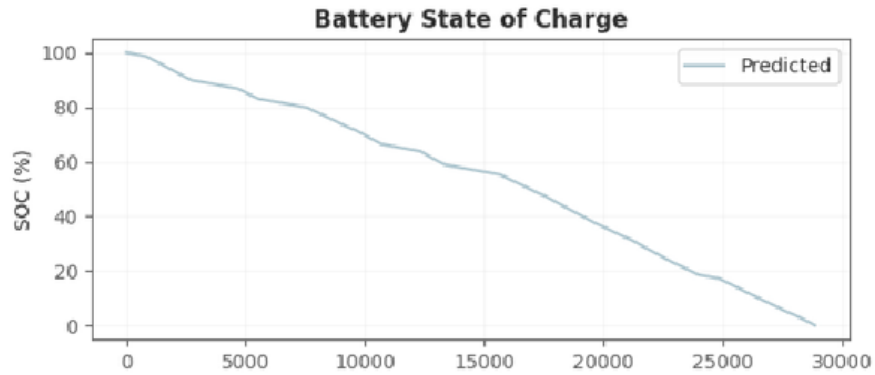
As shown in Figure 13, fluctuating usage patterns lead to non-uniform SOC depletion that reflects transitions between low- and high-power activities. High-intensity scenarios such as video streaming and gaming produce steeper SOC decline, while low-power activities result in slower depletion, yielding a realistic total TTE of approximately 8 hours.

Temperature evolution closely tracks usage intensity: sustained high-load periods induce faster temperature rise and activate thermal throttling, which moderates effective power draw and prevents unrealistically abrupt SOC depletion. Network tail energy, by contrast, responds on a slower time scale and increases only after prolonged network activity.

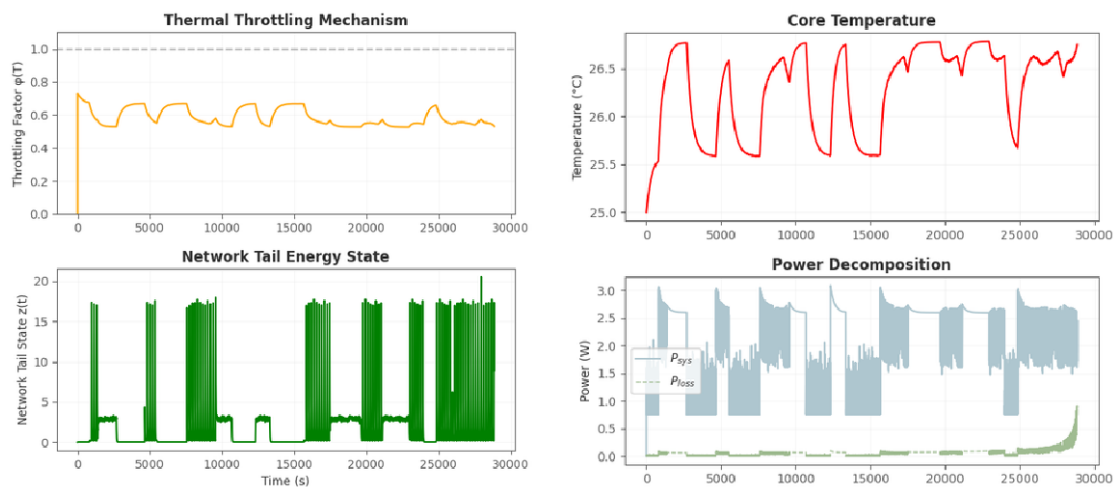
6.4 Adding Gaussian Noise to Inputs

To assess the robustness of the model to input uncertainty, we perform a noise-based sensitivity analysis by injecting Gaussian noise into the input values.

The results (Figure 14) show that SOC predictions are most sensitive to noise in CPU utilization (u_{cpu}), while noise in screen brightness (b) or network activity (x_{net}) has a negligible effect. This reflects the examined usage scenario, in which power consumption is dominated by CPU load and variations in b and x_{net} are relatively weak. Consequently, SOC dynamics are primarily influenced by uncertainty in the dominant power-driving input, indicating that the model remains robust to realistic noise in secondary features.



(a) SOC



(b) Temperature and Power Decomposition

Figure 13: Coupled SOC–thermal–power dynamics under fluctuating usage scenarios. The figure illustrates the evolution of the state of charge (SOC), core temperature, thermal throttling factor, system power decomposition, and network tail energy state under a usage pattern composed of randomly interleaved music, video, and gaming activities.

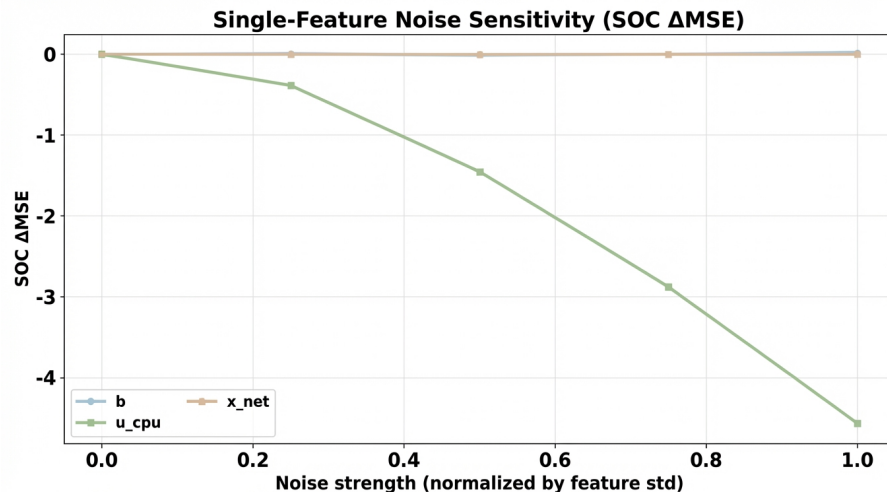


Figure 14: Single-feature noise sensitivity. SOC MSE change versus noise strength for individual input value.

7 Strength and Weakness

7.1 Model Strengths

- The model produces TTE predictions that closely align with observed and plausible SOC trajectories, indicating that the dominant physical drivers of battery depletion are well captured.
- By explicitly modeling power decomposition, thermal feedback, and network tail energy, the framework enables clear identification of the primary contributors to rapid battery drain across usage scenarios.
- Sensitivity analyses with respect to initial SOC, SOH, thermal parameters, and input noise show stable qualitative behavior, confirming the robustness of the model under realistic uncertainties.

7.2 Model Weakness

- Several parameters are selected through heuristic tuning rather than device-specific calibration, which may affect quantitative accuracy for particular hardware configurations.
- The thermal throttling and aging mechanisms are represented using simplified functional forms, abstracting away fine-grained operating system and hardware control policies.
- User behavior within each usage scenario is modeled using stylized patterns, which may not fully capture short-term variability observed in real-world smartphone usage.

8 Conclusion

This paper presents a continuous-time, physics-informed model for smartphone battery behavior that couples power consumption, thermal dynamics, and state-of-charge (SOC) evolution. Simulation results show that time-to-empty (TTE) is primarily determined by usage-dependent power demand and initial charge level, with high-intensity activities such as mobile gaming causing the fastest depletion. The model reproduces first-order SOC depletion trends with reasonable accuracy and remains robust under parameter uncertainty and fluctuating usage patterns. The key innovation lies in explicitly modeling bidirectional thermal feedback, providing a practical basis for understanding and managing smartphone battery life.

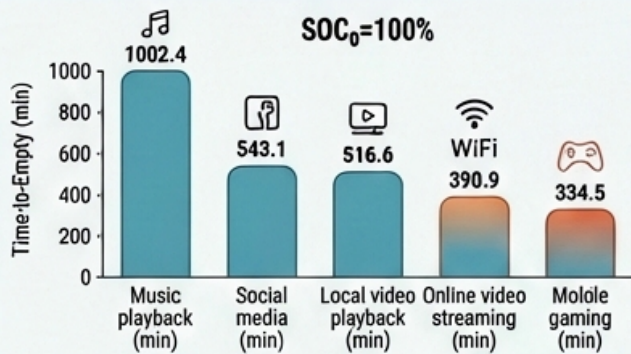
9 Recommendation

See poster below.

A Continuous-Time SOC Model with Thermal Feedback for Smartphone Battery Analysis

We model battery SOC as a coupled power–thermal dynamical system to predict **Time-to-Empty (TTE)** and identify dominant drain drivers across common usage scenarios.

Main Results + Insights



Insights: "who kills the battery" on the scenario

- **[Scenario dominance]** The primary drain driver shifts across scenarios, so "one-size-fits" savings are misleading.
- **[Gaming]** CPU is the dominant driver (51.5%), amplified by thermal stress, sustained high load shortens TTE the most.
- **[Streaming]** Online streaming is a **Screen 49.4% + CPU 38.9%** combo—it's not "just network".
- **[Social & Local video]** Screen dominates: **Social screen 56.3%**, **Local video screen 76.0%** → brightness control 56.3% = lowest radiant power has outsized impact here.
- **[Music]** Surprisingly, **Network 54.9%** due to network tail energy, though the world's for outsized impact here.
- **[SOC scaling]** Initial SOC mainly scales total time, it does not change which analysis which scenario is best/worst.

Practical Recommendations (Actionable + Impact)

Everyday User Actions

Do: Lower screen brightness (or enable adaptive brightness) for social browsing. **Why:** Screen is the largest share in Social (56.3%) and Local video (76.0%). **HIGH**

Do: Avoid long, continuous mobile gaming sessions, take cooldown breaks. **Why:** Gaming is CPU dominant (51.5%) thermal stress, shortening TTE most. **HIGH**

Do: Prefer WiFi when high-speed cellular isn't needed, reduce background syncing. **Why:** Network tail energy can dominate in high scenarios (Music: Network 54.9%). **MED**

Do: When battery is low, avoid high-power tasks (gaming/streaming), switch, switch to low intensity use. **Why:** Under low SOC, "high power per minutes" has the worst cost benefit. **MED-HIGH**

System / OS Policy Ideas

Policy: Thermal-aware performance scaling (smooth throttling) + aggressive brightness governance in screen-dominant apps.

Trigger/Impact: High temps/load → MED-HIGH impact

Network / Thermal Strategy

Strategy: Tail-energy-aware networking (batch downloads, avoid frequent small transfers; prefer WiFi).

Trade-off/Impact: Slight latency vs lower power → MED impact

Model: Continuous-time coupled SOC–power–thermal framework with network tail energy and thermal throttling feedback.

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Report on Use of AI Tools

1. *DeepL Translate* (DeepL SE)
Used to translate paragraphs written in Mandarin into English and to improve linguistic fluency.
2. *OpenAI ChatGPT* (ChatGPT, GPT-5.2)
Query: Provided pre-written content and requested conversion into $\LaTeX$ format; asked for language polishing; submitted code snippets with errors for debugging assistance.
Output: Content rewritten in valid $\LaTeX$ syntax; improved academic language and clear logic; explanations and suggestions for resolving code errors.
3. *Nano Banana* (image-enhancement tool)
Used by uploading draft figures and requesting improvements in visual clarity and presentation quality. The figures numbered 2, 3, 4, and 5 and our poster were processed and visually enhanced using the Nano Banana image optimization tool.
4. *GitHub Copilot* (latest available version, year 2026)
Used for code auto-completion and syntax suggestions during model implementation.